Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/30/2013
PI Name:	Hofmann, Douglas Ph.D.		
Project Title:	Study of Mushy-Zone Development in Dendritic Microstructures with Glass-Forming Eutectic Matrices		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: extended to 10/1/16 (Ed., 7/31/13) NOTE: to be extended past 10/1/2013 per PI (Ed., 7/24/13)		
Key Personnel Changes/Previous PI:	No key personnel changes.		
COI Name (Institution):	Demetriou, Marios (California Institute of Technology) Johnson, William (California Institute of Technology) Rhim, Won-Kyu (California Institute of Technology) Shapiro, Andrew (Jet Propulsion Laboratory)		
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Task Description:	The future materials for NASA spaceflight applications are likely to be high-performance composites. Metallic glass matrix composites, which are metallic glasses reinforced with crystalline dendritic inclusions, were found to impart benchmark properties (see for example, D.C. Hofmann et al. Nature, 451:1085, 2008). The mechanical performance of these composites is understood to be highly dependent on the morphological-feature length scales of the dendritic microstructure. Specifically, it has been demonstrated that the length scales associated with the size and distribution of dendrites has a profound effect on mechanical properties such as ductility, fracture toughness, and fatigue endurance. Understanding and controlling the development of the dendrite microstructure will enable optimization of the mechanical properties and promote a truly unique mechanical performance of these composites. The investigators propose to study the nucleation, growth, directionality, and cooling rate dependence of crystalline dendrites evolving in a glass-forming eutectic matrix. The ground-based research effort will be carried out utilizing laboratories at the Jet Propulsion Laboratory and the California Institute of Technology, including the High Vacuum ElectroStatic Levitator platform (HVESL) at Caltech. The investigators propose a four-component ground-based project consisting of (1) directional solidification experiments utilizing Moving Mushy Zone (MMZ) and Bridgman solidification, (2) morphological characterization of dendrite growth and orientation during "mushy" (or "semi-solid") processing, (3) mechanical testing and development of structure-property relations, and (4) dendrite formation and growth during containerless processing utilizing the HVESL. After completion of a ground-based research program, the investigators will extend the experimentation to flight-based microgravity research aboard the ISS, the aim of which is to study gravity-driven convection arising from density gradients in the microstructure. The flight experiment objectives are to (1) use the Low Gradient Furnace (LGF) to complete Bridgman crystal growth on composites in an attempt to grow dendrites along the translational axis, (2) use the Solidification and Quenching Furnace to coarsen and homogenize dendrite microstructures using mushy-zone processing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of innovative new materials is a fundamental part of future terrestrial and space technologies. These materials have a tremendous benefit towards improving life on Earth, from improving the accessibility and cost of solar power, to making cars safer and stronger, to reducing the mass of aircraft, for example. Innovative technologies typically require new materials tailored for those applications. This drives materials development. Metal matrix composites (MMCs) are a relatively new class of metal alloys that exhibit unique properties which make them desirable for many new technologies. MMCs are typically two-phase alloys consisting of a continuous metal matrix incorporated with a second phase (which can be a metal dendrite, a fiber, a particle, or some other inclusion). They differ from traditional monolithic crystalline metal alloys because the mechanical and physical properties of single-phase alloys are limited by composition and dislocation-based plasticity. By creating a composite phase, MMCs have the ability to combine the beneficial properties of both phases into a single alloy, like strength and toughness, for example. A novel type of MMC has been recently developed that exhibits benchmark combinations of strength and toughness. These alloys were developed by creating a MMC with high-strength metallic glasses (also called amorphous metals) as the matrix phase and a soft body centered cubic (bcc) dendrite as the crystalline phase. These metallic glass matrix composites (MGMCs), which are the focus of the current research effort, have been shown to have the strength of carbides, with the toughness and fatigue limit of stainless steel and the density of titanium. These alloys have the potential for significant impact in high-performance structural applications in areas ranging from defense, to aerospace, to commercial sporting goods. The widespread use of MMC (and MGMCs in particular) requires in-depth research into the processing and fabrication of these multi-phase alloys. For example, semi-solid (or mushy) processing is often required to coarsen the second phases in MMCs and MGMCs to improve the mechanical properties. These additional processing requirements make the fabrication of hardware from MMCs unique compared with traditional metal parts (which are typically machined into a final shape). The NASA-funded technology being developed as part of the current program seeks to understand the development of second phases in MGMCs as a function of processing parameters so that innovative industrial processing techniques can be used to generate hardware. The end result is the potential widespread use of MMC in applications such as biomedical implants or spacecraft shielding.
Task Progress:	During FY13 funding, the PI focused on publishing results from the three-year, ground-based program. The FY11-13 ground-based program was very successful and resulted in the publication of 5 peer-reviewed manuscripts that cited NASA funding, four of which cited the current program as the main source of funding. These include: 1. K. Hozachkov, J. Kolodziejska, W.L. Johnson, D.C. Hofmann, “Effect of cooling rate on the volume fraction of B2 phases in a CuZrAlCo metallic glass matrix composite,” Intermetallics 39, 89-93 (2013) 2. S. Roberts, C. Zachrisson, H. Kozachkov, A. Ullah, A.A. Shapiro, W.L. Johnson, D.C. Hofmann, “Cryogenic Charpy impact testing of metallic glass matrix composites,” Scripta Materialia 66, 5, 284-287 (2012) 3. D. Hofmann, “Bulk metallic glasses and their composites, a brief history of diverging fields,” J. of Materials, 517904 (2013) 4. H. Kozachkov, J. Kolodziejska, S. Roberts, J.Z. Li, W.L. Johnson, D.C. Hofmann, “Study of mushy-zone development in dendritic microstructures with glass-forming eutectic matrices using electrostatic levitation,” ISRN Materials Science, 108363 (2013) 5. (NASA citation but not main) M. Davidson, S. Roberts, G. Castro, R.P. Dillon, A. Kunz, H. Kozachkov, M.D. Dementriou, W.L. Johnson, S. Nutt, D.C. Hofmann, “Investigating amorphous metal composite architectures as spacecraft shielding,” Adv. Eng. Materials, http://dx.doi.org/ (2012). In addition to the publications listed above, there was also a significant amount of progress in both preparation for upcoming flight experiments and the beginning of new ground-based experiments. During FY13 funding, we experimented with fabricating analogue flight samples in using two different techniques. 3 mm rods of bulk metallic glass composites were fabricated through suction casting from the liquid and by infiltrating glass-forming liquid over an insoluble powder. The composite microstructure from these alloys was evaluated and a strategy for fabricating the flight

	samples was developed. Additionally, ground-based research continued on the development of new bulk metallic glass composites, utilizing a new collaboration with UC San Diego. This work involves studying crack propagation in Invar-reinforced Fe-based metallic glass composites. Lastly, we have acquired and are testing brand new equipment for measuring the thermal history of the metallic glass composites during processing. We will be using a high-temperature infrared thermal camera for capturing video of the processing of the composites to gather data on the effect of processing on microstructure and mechanical properties.
Bibliography Type:	Description: (Last Updated: 01/01/2026)
Articles in Peer-reviewed Journals	Hozachkov H, Kolodziejska J, Johnson WL, Hofmann DC. "Effect of cooling rate on the volume fraction of B2 phases in a CuZrAlCo metallic glass matrix composite." Intermetallics. 2013 Aug;39:89-93. http://dx.doi.org/10.1016/j.intermet.2013.03.017 , Aug-2013
Articles in Peer-reviewed Journals	Hofmann DC. "Bulk metallic glasses and their composites: a brief history of diverging fields." Journal of Materials. 2013;2013:517904. http://dx.doi.org/10.1155/2013/517904 (open access journal) , Jan-2013
Articles in Peer-reviewed Journals	Hozachkov H, Kolodziejska J, Roberts S, Li JZ, Johnson WL, Hofmann DC. "Study of mushy-zone development in dendritic microstructures with glass-forming eutectic matrices using electrostatic levitation." ISRN Materials Science. 2013:2013:108363. http://dx.doi.org/10.1155/2013/108363 (open access journal) , May-2013